PRODUCT OVERVIEW

Perfect Hospital Management Solution

Core functionality by user role: front desk, laboratory, pharmacy and management, with the patient journey and the key business rules

VERSION	1.0 — October 2026
AUDIENCE	Hospital owners, administrators, department heads and new staff
COMPANION	A full system documentation (38 pages) covers every rule, the data model and deployment in detail

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1 Introduction

Perfect Hospital Management Solution is a web-based system that runs the daily operations and finances of a hospital in one place: patient registration, OPD consultations, diagnostics and laboratory results, operation theatre cases, in-patient admissions, the in-house pharmacy, and the hospital's accounts.

Staff use it all day from any PC on the hospital network. Each user signs in with their own account and sees only the screens for their role. Receipts, slips and lab reports print on 80 mm thermal printers, and the OPD consultation slip can be printed in English and Urdu. Amounts are in Pakistani Rupees (PKR) and "today" follows the hospital's local day (Asia/Karachi).

What it brings to the hospital

One record per patient

Every patient gets a permanent MR number at registration. Consultations, tests, emergency visits, operations, admissions and pharmacy purchases are all linked to that one record.

One ledger for all money

Every rupee received is recorded once, against exactly one service. Management sees real-time collection, the doctors' share, the hospital's share, expenses and net revenue without manual reconciliation.

Built-in controls

Discounts, free consultations, cancellations and refunds need a written reason. Every change is audit-logged with the user and time. Patients cannot be discharged while money is owed.

Pharmacy with real stock control

Batch and expiry tracking, first-expiry-first-out sales, in-patient credit, returns, stock counts, a complete stock history and an end-of-day cash closing.

Modules

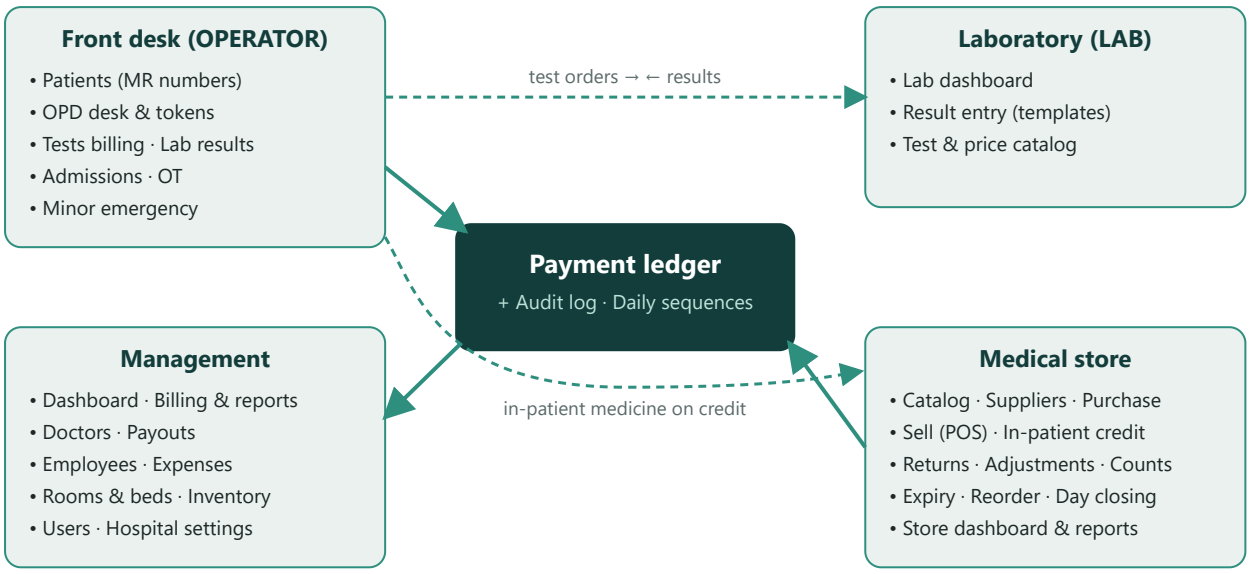


Figure 1 — Modules and how they connect. All money flows into one Payment ledger that management reports on.

2 Users and roles

Every user has exactly one of four roles. The role decides which screens appear in the sidebar. More importantly, the server checks the role on every action, so hiding a link is never the only protection.

Role	Who	What they do
Operator	Front desk / reception	Registers patients; issues OPD tokens and slips; bills diagnostic tests; hands out lab reports; records emergency visits; runs OT cases and in-patient admissions, including payments, invoices and discharge.
Lab	Laboratory technicians	Maintains the test and price list with result templates; enters laboratory results.
Medical Store	Pharmacy staff	Runs the pharmacy: medicines, suppliers, purchases, sales, in-patient medicine, returns, stock control, day closing and reports.
Management	Owners / administrators	Financial dashboard and reports; doctors and their shares; payouts, salaries and expenses; rooms and beds; general inventory; user accounts; hospital settings. Can also use the pharmacy functions.

Screens per role

Role	Sidebar
Operator	Dashboard · Patients · OPD desk · Tests · Lab results · Admissions · Operation theater · Minor emergency
Lab	Lab dashboard · Upload tests (result entry) · Test & Price Catalog
Medical Store	Store dashboard · Medicine catalog · Suppliers · Supplier payments · Purchase · Reorder list · Sell medicine · In-patients · Day closing · Stock adjustments · Stock count · Stock history · Expiry tracker · Sale returns · Purchase returns · Reports
Management	Dashboard · Doctors · Billing & reports · Payouts · Employees · Expenses · Hospital settings · Inventory · Rooms & beds · User access

Access at a glance

Function	Operator	Lab	Store	Mgmt
Register and edit patients	✓	—	—	✓
OPD, emergency, OT, admissions	✓	—	—	✓
Bill, cancel or refund tests	✓	—	—	✓
View test results	✓	✓	—	✓
Enter laboratory results	—	✓	—	✓
Test & price catalog	—	✓	—	—
Pharmacy (all functions)	—	—	✓	✓
Finance, doctors, staff, beds, users, settings	—	—	—	✓

Accounts

- Management creates accounts. New and reset accounts must set their own password (at least 8 characters) before using anything.
- Sessions last 8 hours. Five wrong passwords lock that username for 15 minutes.
- Accounts are never deleted, only deactivated, so history always shows who did what.

3 The patient journey

A patient is registered once. From there they may see a doctor in OPD, have tests, come through minor emergency, have an operation, or be admitted. Simple visits are paid on the spot. Operations and admissions keep a running balance that must be cleared before discharge.

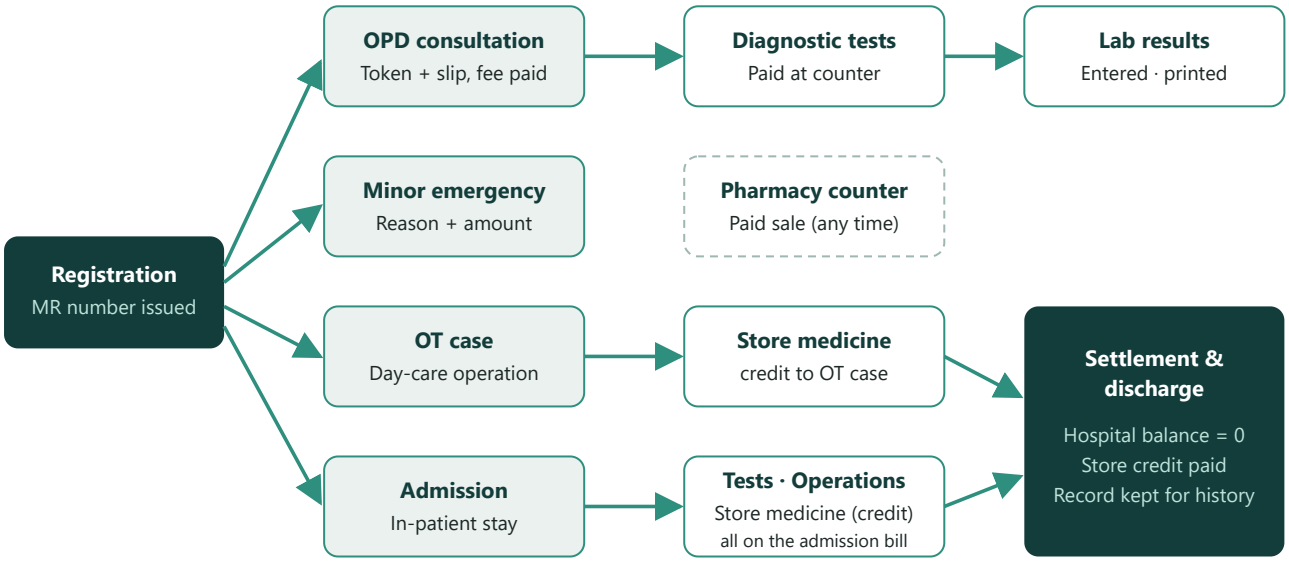


Figure 2 — Patient journeys. OPD, emergency and counter pharmacy are paid on the spot. OT cases and admissions run a balance that must be cleared before discharge.

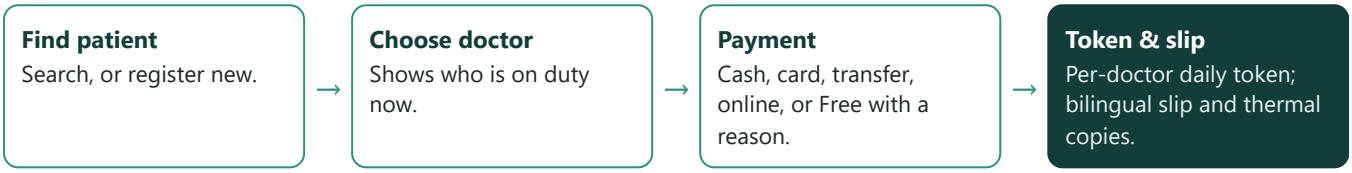
Encounter	Payment	Closes when
OPD visit	Fee paid when the slip is issued (or waived with a reason)	Immediately
Diagnostic tests	OPD: paid at the counter. In-patient: added to the admission bill	Results entered and handed over
Minor emergency	Paid on the spot	Immediately
OT case	Full, advance or due; instalments allowed	Discharge: nothing owed to the hospital or pharmacy
Admission	Advances, part-payments, final settlement	Discharge: balance exactly zero and pharmacy paid
Pharmacy sale	Paid on the spot (in-patients: on credit)	Immediately (returns possible)

4 Front desk (Operator)

4.1 Patients

- Register with name, father's name and gender (required), plus CNIC, mobile, date of birth and address. CNIC and mobile are auto-formatted and must be unique, which prevents duplicate records.
- An **MR number** (e.g. `MR-2026-000412`) is issued automatically and never changes.
- Search by any part of the MR number, name, CNIC or mobile. The patient history shows all OPD visits, test reports and every edit made to the record.

4.2 OPD desk



- Tokens restart at 1 for each doctor every day, and two desks never get the same number.
- The doctor's consultation slip is printed in English and Urdu with space for the handwritten prescription.
- A visit can be deleted only with a reason, and only before any tests are billed under it.

4.3 Diagnostic tests and lab results

- Bill Laboratory, X-Ray, Ultrasound, ECG or ECO tests against the patient's OPD number. Prices come from the lab's catalog.
- Discounts need a reason. Cancellations and refunds need a reason and are removed from income.
- The **Lab results** screen refreshes itself. The desk gets a notification the moment the lab finishes a test, then prints the report on 80 mm paper.

4.4 Minor emergency

Select the patient, enter the reason and amount, take payment and print the receipt. Each visit gets a daily token and number.

4.5 Operation Theatre

- Create a case with patient, surgeon, room, procedure and fees (surgeon, theatre, anaesthesia, room, hospital). The payment method must be chosen explicitly: full payment, an advance, or **Due**.
- Add consumables during the case, collect the balance in any number of instalments, and print the invoice at any time.
- Medicine comes from the pharmacy on credit to the case. **Discharge is blocked** until the hospital balance is zero and the pharmacy has been paid.

4.6 Admissions (in-patients)

Admission type	Allowed beds
Medical / Surgical	Private room or Ward
Critical	Critical care
NICU	NICU (newborns can be registered right on the admission form)

- **Admit:** choose the patient, type, attending doctor and a free bed (or a manually entered room). Daily room, doctor-visit and technician rates are filled in and can be adjusted. An advance can be taken.
- **Daily charges** are added automatically. A stay under 24 hours is one day; after that, every calendar date is charged, never twice. If the doctor did not come on a day, the operator unticks it and that day's doctor fee is dropped.
- **During the stay:** add extra charges, add an operation, order tests (no counter payment; they go on the bill), change bed, doctor or rates, and collect advances. The pharmacy issues medicine on credit.
- The bill shows the **hospital part** (room, doctor visits, technician, extras, operations, tests) separately from the **pharmacy part** (medicine for ward, OT and home).

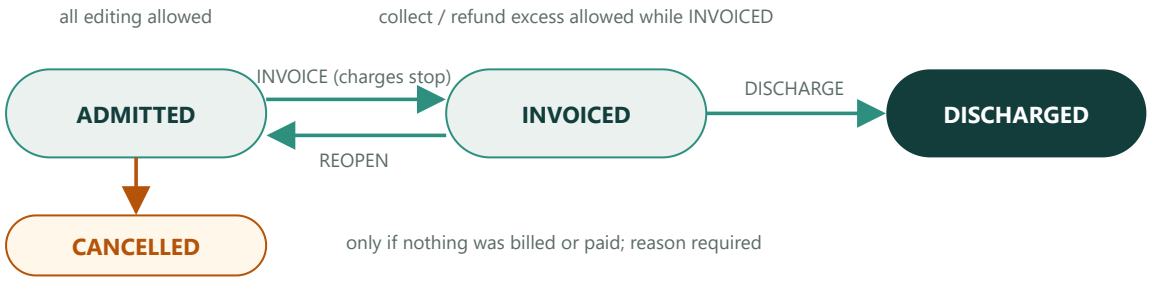


Figure 3 — Admission statuses.

Discharge requires

The final invoice (charges stop at that moment), a hospital balance of **exactly zero** (any excess advance is handed back first), and the pharmacy bill marked paid. An admission made by mistake can be cancelled with a reason, but only if nothing was billed or paid.

4.7 Operator dashboard

Today only: OPD patients and last token per doctor, doctors on duty, tests and results returned, emergency visits, admissions (admitted, discharged, awaiting discharge), open OT cases, and the desk's collection by service and payment method, including the signed-in user's own share.

5 Laboratory (Lab)

5.1 Test & price catalog

- Tests per module (Laboratory, X-Ray, Ultrasound, ECG, ECO) with price and default remarks.
- Laboratory tests have a **result template**: section headings, numeric values (unit, normal range, optional second unit), positive/negative values, and free text.
- About **60 ready-made templates** (CBC, LFT, RFT, lipid profile, thyroid, HbA1c, urine and stool R/E, dengue, hepatitis, hormones and more) load in one click and can be adjusted to the lab's own ranges.

5.2 Entering results

Queue
New test orders appear automatically, from OPD and from admitted patients.

→

Enter values
Checked: numbers must be numeric, positive/negative must be one of the two.

→

Auto-flag
High / Low against the range; Positive flagged abnormal.

→

Save
Report is fixed; front desk is notified to print.

- Each saved report keeps its own copy of names, units and ranges, so later template changes never alter issued reports.
- The Lab role enters Laboratory results only; imaging and ECG results need the patient present.
- The lab dashboard shows pending tests, completed tests, receipts, tests ordered and cash collected, with trends and a test-type breakdown.

6 Pharmacy (Medical Store)

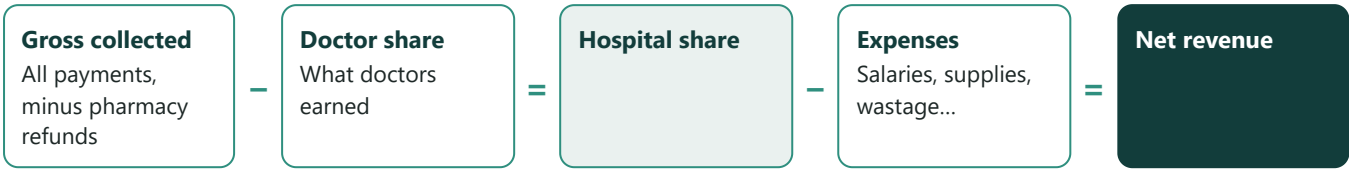
Area	Functionality
Medicine catalog	Name, generic name, category, base unit, sale price, low-stock level, and **packaging levels** (e.g. 1 Box = 10 Strips, 1 Carton = 12 Boxes). Quantities can be entered in any pack size.
Suppliers & purchasing	Each purchase line creates a separate **batch** with batch number, expiry and total cost; the system works out the cost per unit. Supplier payments and the balance owed are tracked.
Selling (POS)	Patient or walk-in customer, optional discount with reason, cash/card/transfer/online, 80 mm receipt with token. Stock is drawn **first-expiry-first-out**; expired stock is never sold.
In-patient medicine	Issued on credit to an admitted or OT patient, marked as Ward, OT or Home medicine. When the family pays, the store clicks **Mark paid**; only then can the patient be discharged.
Returns	Sale returns refund what was actually paid (discounts pro-rated) and either restock or quarantine the stock. Purchase returns go back to the supplier as cash or credit.
Stock control	Adjustments for damage or expiry (recorded as wastage at cost), physical stock counts, expiry tracker, reorder list, and a full stock history of every movement.
Day closing	Expected cash (cash sales − cash refunds) compared with counted cash; the difference and a note are saved.
Dashboard & reports	Today's revenue, low-stock / expiring / expired counts, revenue trends; date-range reports with profit, top sellers and supplier purchases, exportable to Excel.

Stock can never go negative

Sales, adjustments and returns are all checked against each batch's remaining quantity, and every movement is logged with the before and after quantity.

7 Management

7.1 Financial dashboard and reports



- Any period (today, week, month, custom) compared with the previous one; revenue by service; revenue trend; patients seen; in-patients and OT right now; what doctors are owed.
- **Billing & reports:** every transaction, expense and doctor payout for a date range, exportable to Excel or printed as PDF.

7.2 Doctors and payouts

Income stream	Doctor's share
OPD consultations	Hospital keeps a percentage, or a fixed amount per patient; the doctor gets the rest
OT surgeon fee	
Admission visits	A separate split per visited day
Tests referred	A referral percentage per module (Lab, X-Ray, Ultrasound, ECG, ECO)

The Payouts page shows each doctor's earnings for the period and a running **balance due** (all-time earned minus all-time paid). Paying a doctor settles that balance; it is not counted as a hospital expense, because the doctor's share was already taken off the gross.

7.3 Employees, salaries and expenses

- Each employee has a monthly salary and a payday. The system shows who is upcoming, due today or overdue, and prorates a new joiner's first salary.
- Salary payments can include an incentive or a deduction. A salary cannot be paid before its payday, or twice for the same payday.
- Expenses are recorded by category (general, department, salary). Inventory purchases, supplier payments and pharmacy wastage are posted automatically.

7.4 Setup and administration

Screen	What is configured
Hospital settings	Name and address (English and Urdu), phone, email, logo: shown on the login screen and printouts
Doctors	Names and qualifications (English/Urdu), weekly schedule, consultation fee, hospital split, referral %, admission fee and split
Rooms & beds	Beds with category, daily rate, per room / per bed; technician fee per admission type
Inventory	OT tools and consumables, with restocks recorded as expenses

Screen	What is configured
User access	Create accounts, assign roles, reset passwords, deactivate

8 Key business rules

#	Rule
1	A patient is registered once, gets a permanent MR number, and cannot be duplicated by CNIC or mobile.
2	OPD tokens restart daily per doctor and are never duplicated, even when two desks issue at once.
3	Every discount, free consultation, cancellation and refund requires a written reason.
4	Every payment belongs to exactly one service, so nothing is counted twice.
5	Prices and lab templates are copied onto receipts and reports; later changes never rewrite history.
6	OT and admission patients cannot be discharged while anything is owed to the hospital or the pharmacy.
7	An admitted bed cannot be given to another patient; each admission type may use only its bed categories.
8	Admission days are charged once per calendar date (under 24 hours = 1 day); no doctor fee on days the doctor did not come.
9	After the final invoice, charges stop and only settlement, refund of excess, reopening or discharge are allowed.
10	Medicine is sold first-expiry-first-out; expired stock is never sold; stock never goes negative.
11	In-patient medicine is income only when the pharmacy marks it paid; returns before payment just lower the bill.
12	Doctor payouts settle what is owed and are not hospital expenses.
13	Salaries cannot be paid early or twice; a new joiner's first salary is prorated.
14	Users are never deleted; every change is recorded in the audit log with the user and time.

9 Security and platform

Security

Passwords are stored only as salted hashes. Sessions are secure-cookie based, expire after 8 hours, and require HTTPS. Logins are throttled. Every action is checked against the user's role on the server, and every change is audit-logged.

Reliability

Multi-step saves are all-or-nothing database transactions. Numbering is safe when several users save at once. Daily charges need no scheduler, and live screens refresh themselves.

Technology

Next.js and React web application with TypeScript; MySQL/MariaDB database; runs on standard cPanel Node.js hosting or a local server.

Operations

A health-check address, regular database backups with tested restores, and thermal 80 mm printing for slips, receipts and reports.